MSC ARTIFICIAL INTELLIGENCE DISSERTATION

# Learning to Select Actions for Complex Hand Tasks

A comparative study of visuomotor policies and latent world models

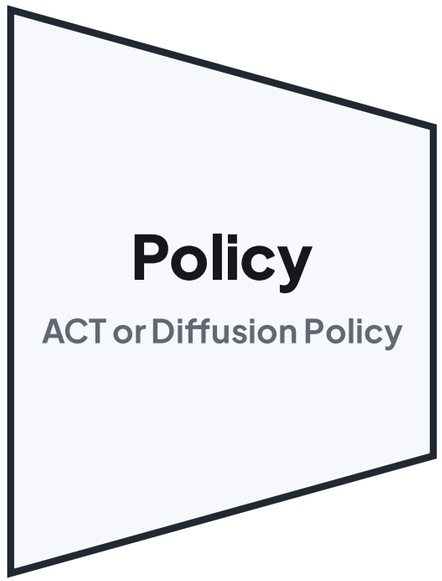
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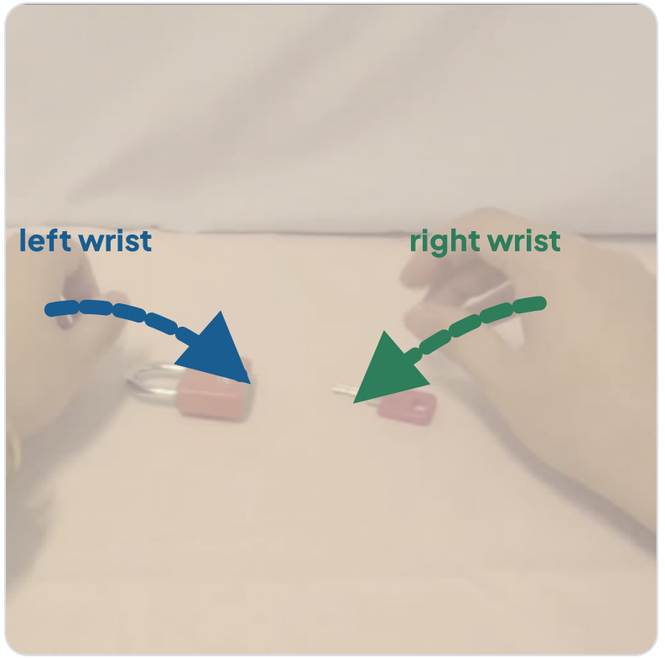
# What is a policy?

A policy is a model that learns from demonstrations (human, robot or simulated) to turn the current scene into a chunk of future actions.



Input: the current scene

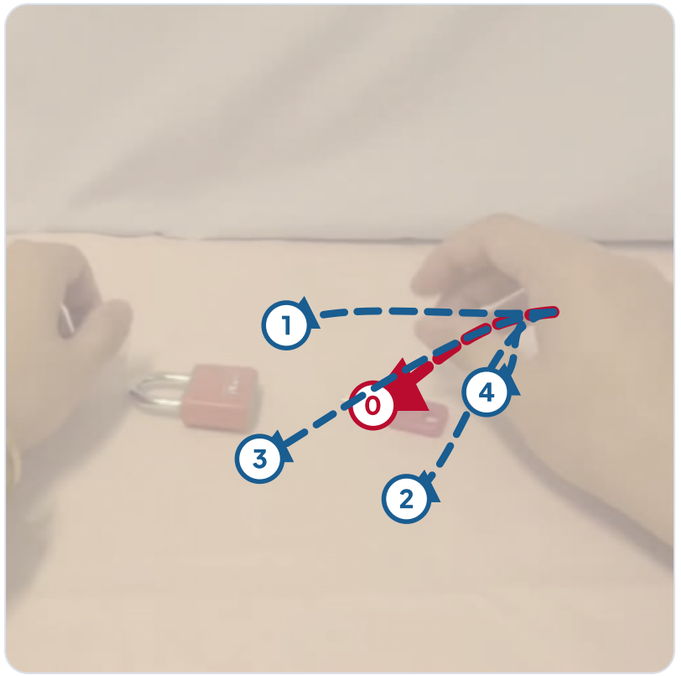



Output: the next 16 frames of motion for both wrists

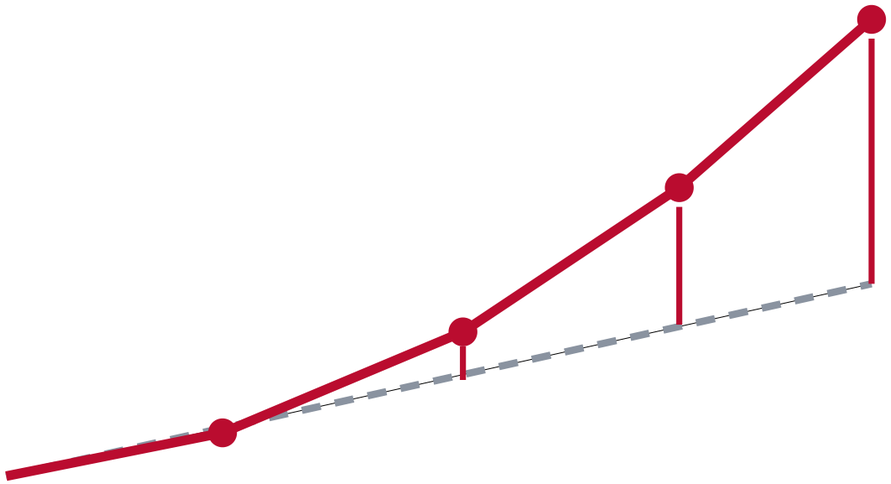
WHERE A POLICY FALLS SHORT

# A policy can produce many motions, but commits to one




Different latent or noise samples give different candidates. Candidate 0 is the one the policy prefers by its own criterion.

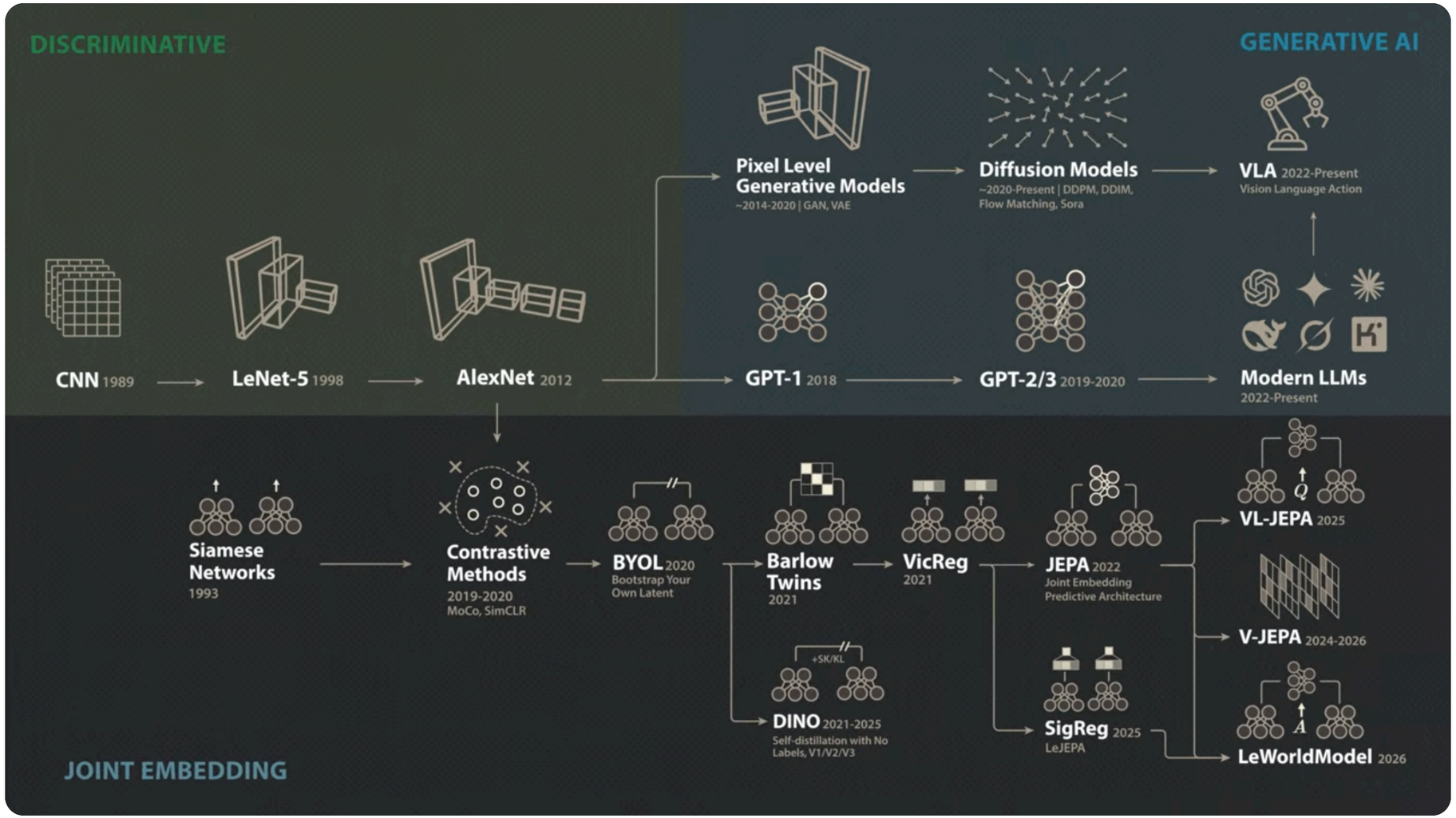
ERRORS COMPOUND



expert motion (dashed) policy rollout

When a suboptimal candidate is chosen, errors compound and can lead to an unsuccessful or dangerous state.

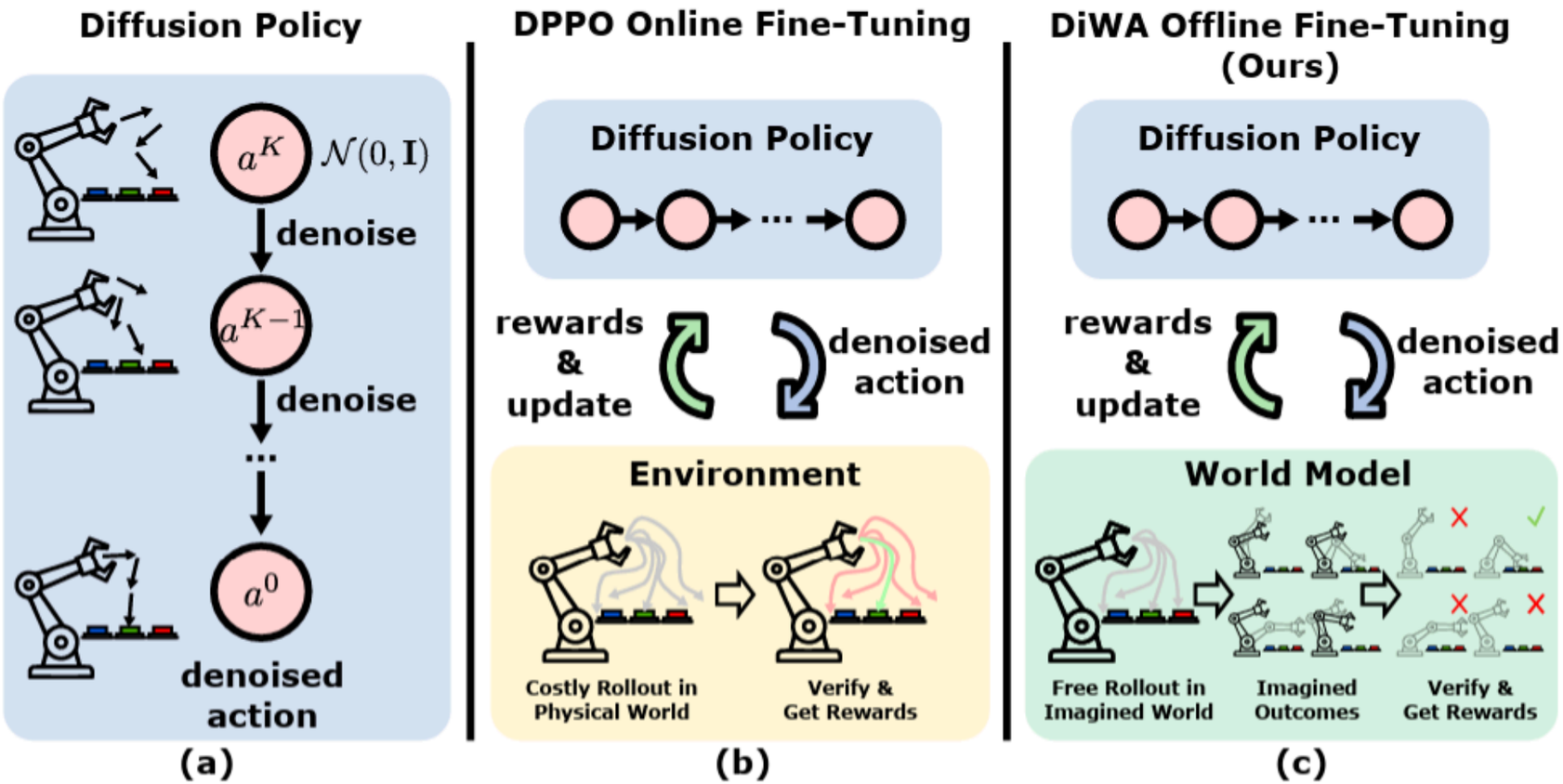
# Generate the action directly, or predict the future and choose




Screenshot from Welch Labs, Yann LeCun's $1B Bet Against LLMs (2026).

THE CLOSEST PRIOR WORK

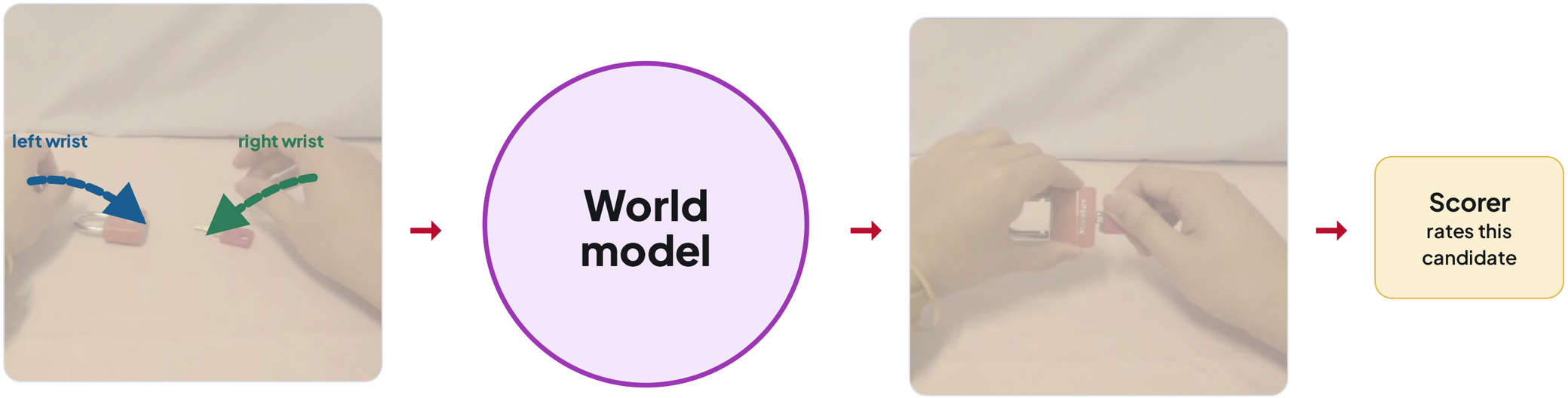
# DiWA uses a world model to retrain the policy entirely offline




Chandra et al. (2025). (a) Imitation learning is limited by offline data. (b) DPPO finetunes with costly online interaction. (c) DiWA finetunes offline through imagined rollouts in a learned world model.

THE PREDICTION UNIT

# A world model predicts what happens next, and a scorer turns that into a choice



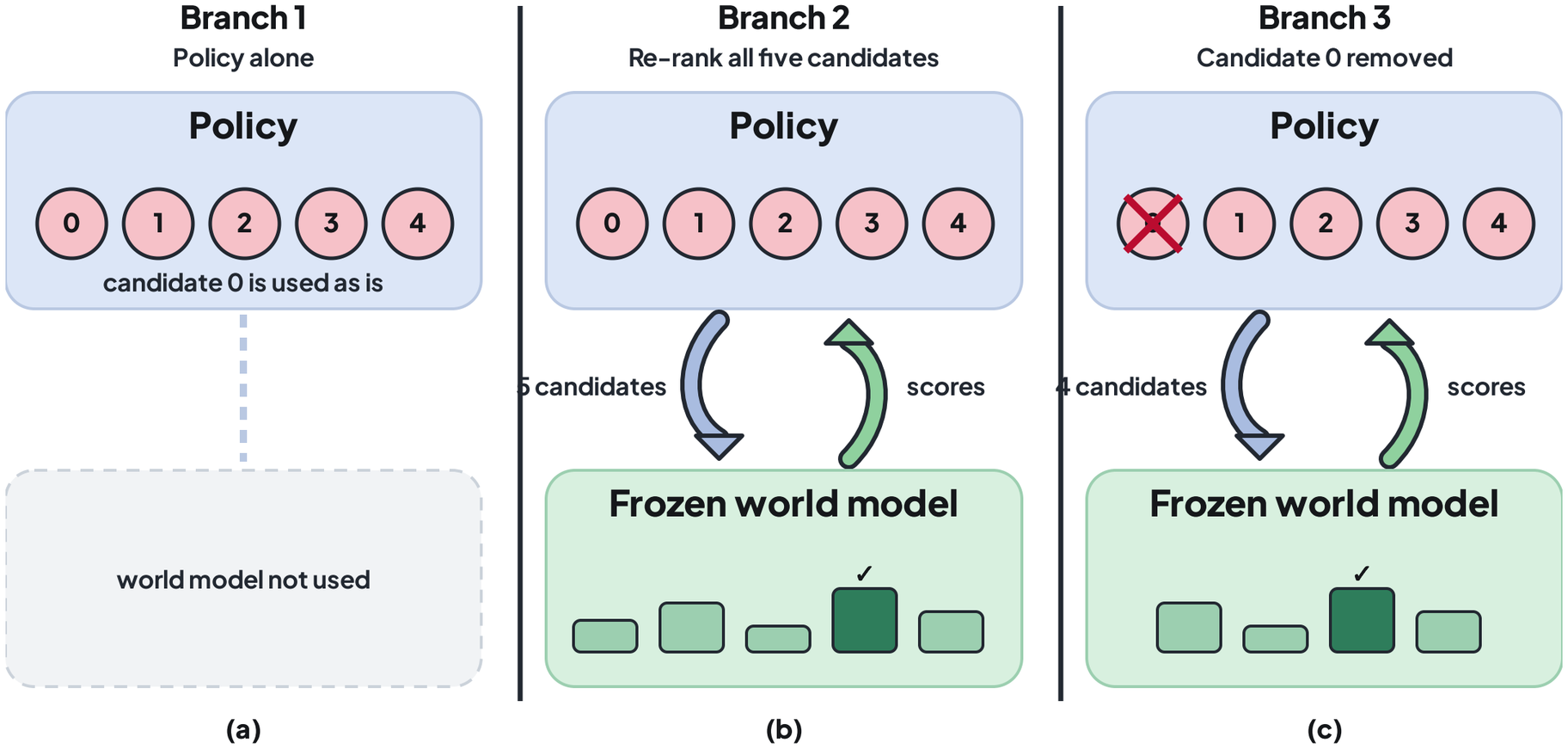

Input: the scene and one candidate motion

Predicted scene afterwards

The world model does not decide what to do. A prediction is not a decision, so a scorer must turn each predicted future into a score. A policy alone also needs a rule to pick, which is why selection is the question. Illustrative frames: the real prediction is made in a compact latent space, not in pixels.

THE VARIANT EXPLORED HERE

# Same policy, same candidates, a frozen world model as the judge




No rewards and no update, unlike DiWA: the policy is never retrained. Bars show the world model's score for each candidate, and the highest score is chosen. Branch 3 removes the policy's favourite so the world model must find better candidates.

FROM THE GAP TO THE QUESTION

# Can a frozen world model help just by choosing?

RESEARCH QUESTION

Can a frozen world model, acting only as the final decision stage over candidates a policy already proposes, improve the selected motion chunk?

## Cheap

The policy is never retrained. Selection can sit on top of a policy that already exists.

## Clean

The candidates are fixed, so any change comes from the world model's judgement and nothing else.

## Safe

The world model never generates motion. It only ranks motions the policy already proposed.

WHAT THE STUDY DOES

# Three objectives: build, compare, diagnose

### Build

A controlled offline benchmark that separates the policy's preference from the world model's.

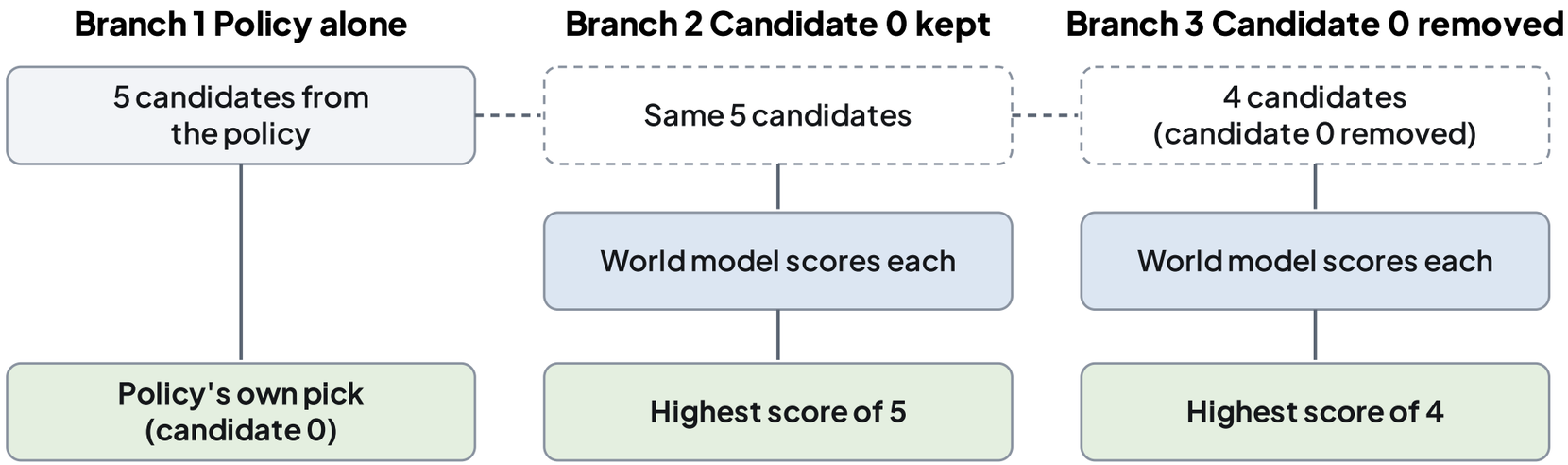
### Compare

Five world-model families and two policies, under matched local training exposure.

### Diagnose

When selection helps, and a likely explanation for each case where it does not.

# One pool of candidates, three ways to choose




**7,607**
HELD-OUT WINDOWS

**16 to 16**
FRAMES IN, FRAMES OUT

**18-D**
BIMANUAL WRIST MOTION

**K = 5**
CANDIDATES PER WINDOW

**3 seeds**
POOLED RESULTS

Data is EgoDex egocentric video. Error is measured against the expert motion as translation (metres, per axis) and rotation (degrees). Branch 1 is the baseline everything is compared with.

# Where the candidates come from changes what the result means

**Diffusion Policy**

5 noise seeds → 5 distinct motions → World model ranks them

Independent pool, clean measurement

**ACT**

5 latent samples → 1 identical motion (posterior collapse) → CEMP search scored by the world model

Scorer shapes its own pool before selecting

**WHAT HAPPENED TO ACT**

Four retraining attempts changed the KL weight, annealing, observation dropout and free bits. The latent was still ignored, and 7 of 12 checkpoints collapsed across seeds.

**WHAT THAT MEANS**

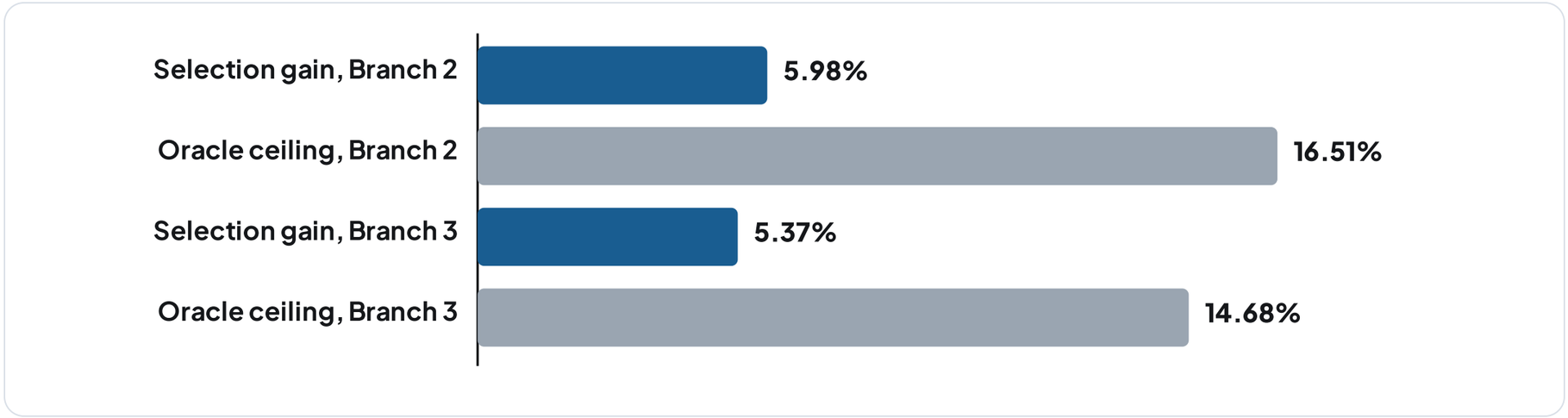
ACT results measure the world model and planner together. Diffusion Policy has no such coupling, so it carries the main claim.

# Five world-model families, plus two controls

| MODEL | WHAT IT IS | ROLE IN THIS STUDY |
|---|---|---|
| **V-JEPA2-AC** | Frozen video backbone pretrained on over a million hours, with an action-conditioned head trained locally | Main candidate scorer |
| **DexWM** | Dexterous-manipulation world model on a DINOv2 encoder | Trained at a fixed budget, one seed |
| **LeWM** | Compact encoder and predictor trained from scratch | Two scorer configurations |
| **DINO-WM, JEPA-WM** | Official checkpoints with a 7-D action interface | Validation pilot through an adapter |
| **No-AC, PointWorld proxy** | Same backbone with the action input removed or weakened | Controls for the mechanism |

Every locally trained model receives the same exposure of 100,000 items. Externally pretrained backbones still carry training that budget does not offset, so results are read model by model.

# Ranking Diffusion Policy candidates with V-JEPA2-AC cuts translation error by 5.98%




| -0.96° | about 36% | 49.9 / 23.2 / 27.0 | 0.00% |
|---|---|---|---|
| ROTATION ERROR | OF ORACLE HEADROOM | WIN / TIE / LOSS, % | WITHOUT ACTION INPUT |

Pooled across three training seeds, 95% CI [5.43, 6.47]. No retraining. The oracle ceiling is what a perfect ground-truth selector could remove from the same five candidates. Translation error is root-mean-square per axis.

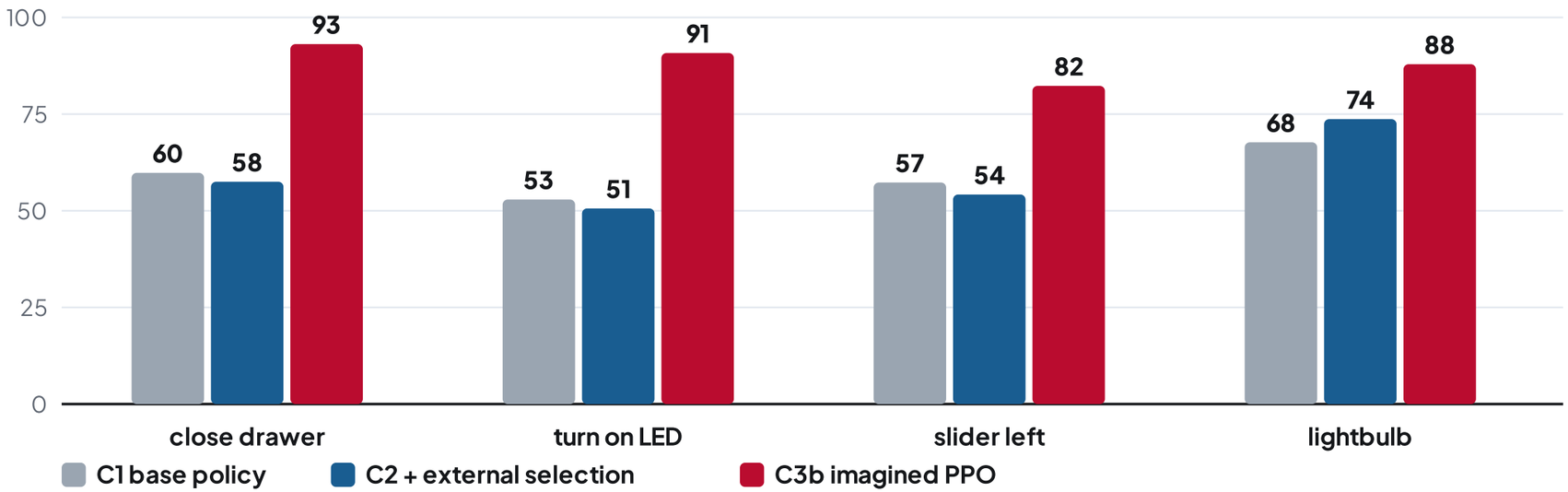
# The other routes did not reproduce it, and one result needs care

**LEWM: THE LARGEST NUMBER, NOT THE BEST EVIDENCE**

LeWM shows a 10.1% translation reduction, but its score does not predict which candidate is better (rank correlation −0.51 on the full test set). Disabling its dynamics model gives almost the same gain. It is reported as unvalidated and held out of the headline.

| ROUTE | OUTCOME | LIKELY EXPLANATION |
|---|---|---|
| **DexWM** | +0.57% worse one seed | Predictor trained at a fixed budget, published checkpoint unavailable |
| **DINO-WM, JEPA-WM** | Latent score only | Adapter squeezes 18-D actions into a 7-D interface |
| **PointWorld proxy** | +0.27% not significant | Weakened action input, a control by design |
| **ACT with CEMP** | −0.58% small | Scorer shapes its own pool, read as a unit |

# Finetuning inside a world model beats selecting with one




**SELECTION (C2 VS C1)**

Moves tasks by −3.1 to +6.1 points, with overlapping intervals.

**FINETUNING (C3B VS C2)**

Adds 14 to 40 points on every task, about an order of magnitude more.

Task success (%), four CALVIN tasks, DiWA's own checkpoints, pooled across three seeds.

# Selection helped when three conditions held together

### 1 Diverse pool

There must be real alternatives to choose between. ACT's collapsed pool offered none until search restored it.

### 2 Action information

The scorer must see the action. Removing it gave exactly 0.00%.

### 3 In-support candidates

Candidates must lie within what the scorer was trained on. Adapter pilots improved a latent score and not the trajectory.

**WHY FINETUNING WINS**

DiWA's imagined-PPO finetuning meets all three conditions by construction, because its reward signal is what the policy is trained against. External selection has no such guarantee.

This is a pattern supported by the tested configurations, not a general law.

# What this does and does not show

**LIMITS**

- Offline trajectory error, not robot task success
- One dataset and one positive pairing
- Matched exposure, not equal pretraining
- LeWM's gain is unexplained
- Selection worsens the easiest fifth of windows

**NEXT STEPS**

- Closed-loop execution, starting with CALVIN-style tasks
- A second, independent world model as selector over a finetuned policy
- Native 18-D finetuning of DINO-WM and JEPA-WM
- DexWM at its published scale
- A diversity mechanism for ACT

Contribution in one sentence. A controlled offline benchmark and protocol showing a frozen world model can improve which of a policy's own proposed motions is chosen, for one tested pairing, without retraining and without any claim yet about real-robot success.

THANK YOU FOR LISTENING

# Questions

Backup slides follow, covering the AI-use workflow, how each method was chosen, LeWM's gate, seeds, pool size, and where selection fails.

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# How AI was used in this work

## NotebookLM

Queried paper sets for related work and citations, then each source was skimmed manually.

## Claude skills

Query, draft and polish workflows, with my own review and edits.

## Gemini

Paper discovery and critical review of chapters against a rubric.

## Coding agents

Wrote most of the code under my specification, then tested and audited.

### WHERE THE DECISIONS SAT

Research question, benchmark design, which methods to adopt, what counted as a valid result, and every claim in the dissertation.

### HOW CORRECTNESS WAS CHECKED

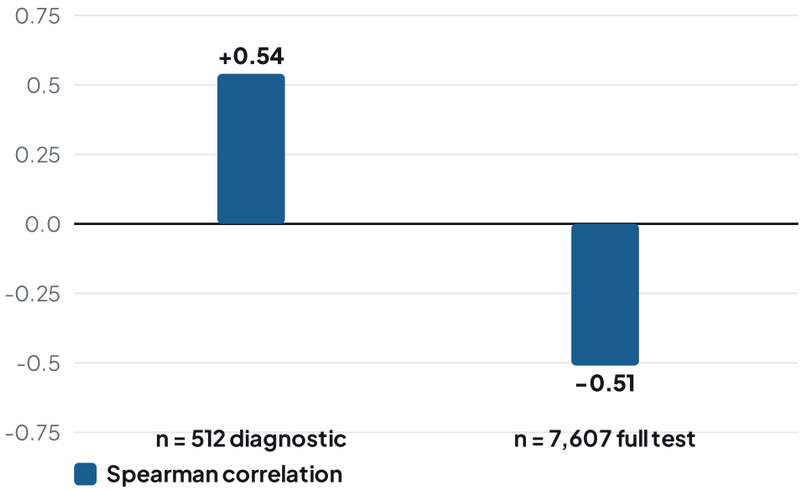
Separate audit passes and re-runs found real defects, including a DDIM sampler indexing error, a training-exposure undershoot, and a DiWA configuration mistake. All were fixed before reporting.

Used most in the literature review and methodology, for example to find established reference methods for handling ACT and LeWM without inventing new verification tests.

# Which published work anchored each method choice

| CHOICE | ANCHOR | STATUS |
|---|---|---|
| Policy-guided candidate search (CEMP) | Cross-entropy method, initialised from the policy output, inspired by TD-MPC2 | Adapted, not a published recipe |
| Posterior-collapse fixes for ACT | KL annealing (Fu et al.), free bits (Kingma et al.), observation dropout | All tried, all failed |
| Diffusion sampling | DDIM (Song et al.), DP-T architecture (Chi et al.) | Implemented, sampler bug fixed |
| Action-conditioned scorer | V-JEPA2-AC (Assran et al.) | Head trained locally |
| Finetuning comparison | DiWA (Chandra et al.) on CALVIN, own checkpoints | Reproduced, secondary study |
| Pool size K = 5 | No literature anchor | Compute choice, ranking stable K = 2 to 20 |
| Matched exposure 100,000 items | Compute ceiling, loss still falling | Not a convergence claim |

# LeWM passes its ranking gate on 512 windows and fails it on 7,607




**WHAT IS AND IS NOT ESTABLISHED**

- The gain is real, −10.1% translation, intervals exclude zero
- The reported configuration never invokes LeWM's learned dynamics
- Its score does not predict which candidate is better at full scale
- Reweighting terms to pass the gate was rejected as overfitting the gate

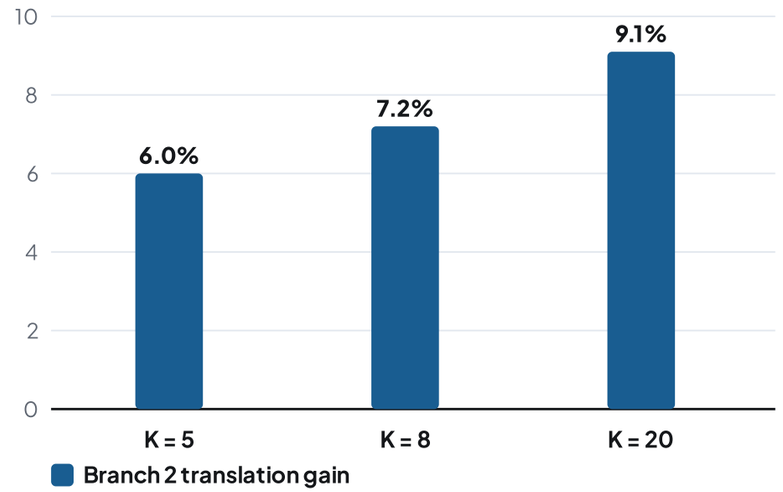
Spearman correlation between score margin and true translation improvement. The 512-window figure was +0.537, the full test set gave −0.51.

# The result holds at every seed, and the controls behave as designed

| | SEED 42 | SEED 43 | SEED 44 |
|---|---|---|---|
| Baseline translation error | 0.1142 | 0.1178 | 0.1191 |
| **V-JEPA2-AC, Branch 2** | -6.41% | -6.09% | -5.46% |
| V-JEPA2 without action input | 0.00% | 0.00% | 0.00% |
| PointWorld proxy | +0.50% | +0.23% | +0.07% |
| LeWM rollout (unvalidated) | -10.10% | -10.06% | -10.12% |

Three seeds are a basic robustness check, not proof of stability. DexWM is single-seed by a cost decision of roughly 121 extra GPU-hours.

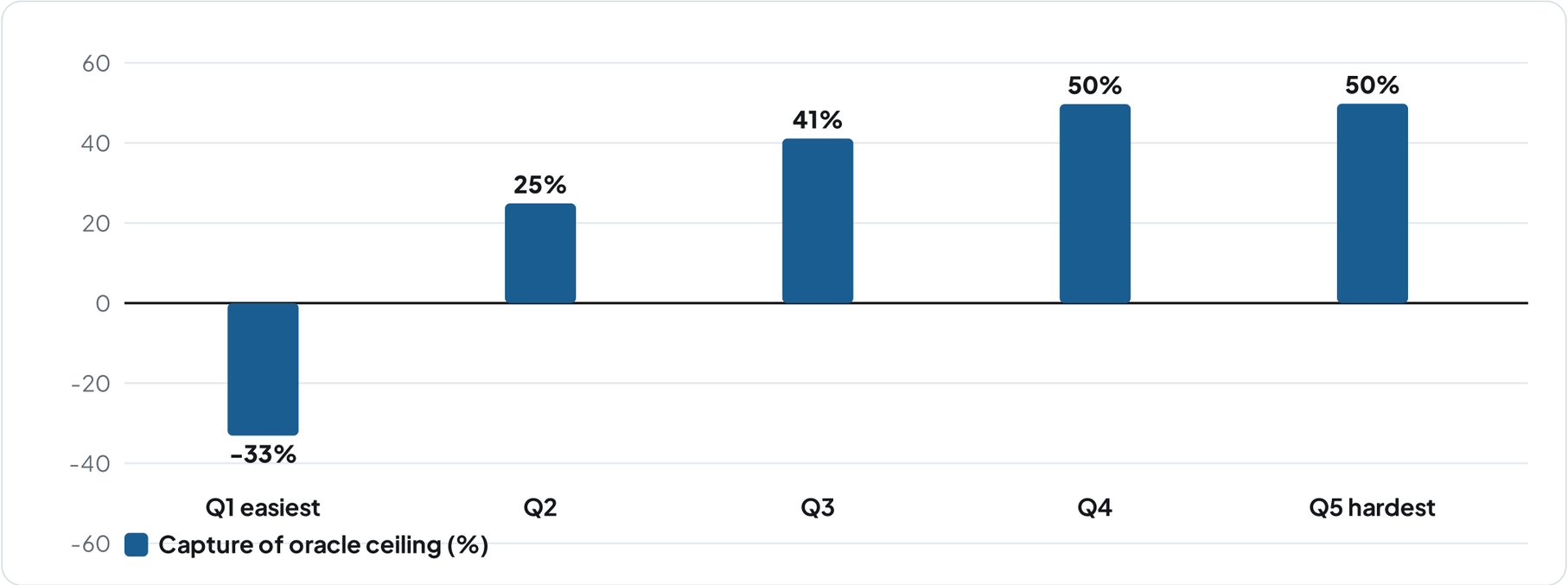
# More candidates buy more headroom, not better selection




**WHAT THE SWEEP SHOWS**

- Scorer ranking identical at every K from 2 to 20
- Gain grows steadily with K and does not plateau
- Capture of the oracle ceiling stays near 40%
- K = 5 was a compute choice, with no literature anchor

# On the easiest fifth of windows, selection makes things worse




Windows sorted by baseline error, K = 8, Branch 2. Translation error rises 5.81% in Q1. Confidence gating does not help, because the score gap has a correlation of -0.002 with baseline error. A state-change-targeted windowing removes most of the reversal.

# Four retraining attempts, the same collapse

| RUN | KL WEIGHT | CYCLICAL KL | OBSERVATION DROPOUT | FREE BITS | PAIRWISE DIFFERENCE |
|---|---|---|---|---|---|
| v1 | 10 | no | 0% | no | indistinguishable from zero |
| v2 | 100 | no | 0% | no | indistinguishable from zero |
| v3 | 100 | yes | 30% | no | indistinguishable from zero |
| v4 | 100 | yes | 0% | 0.5 nats/dim | indistinguishable from zero |

Collapse is seed-dependent. Under the default configuration 1 of 3 seeds collapsed, and 7 of 12 checkpoints collapsed across all configurations. EgoDex observations are informative enough that the decoder finds a loss minimum that ignores the latent. A repair would need an architectural change such as discrete latents or a diversity head.

# How results were measured

### ERROR METRICS

- Translation, root-mean-square per axis, not Euclidean distance
- Rotation, mean geodesic angle in degrees
- Both averaged over the full 16-step chunk
- Change reported relative to candidate 0

### STATISTICS

- 95% episode-clustered bootstrap, 2,000 resamples, pooled over seeds
- Oracle ceiling, best candidate in the same pool
- Ranking-validity gate, per-window Spearman correlation
- Win, tie, loss against candidate 0

Test windows are fixed, candidate pools frozen, future-goal access blocked, and scorer support restricted to training data.

# What is disclosed about test-set use

- The Branch 2 margin threshold's final value came from a disjoint validation split.
- An earlier, superseded calibration touched the test set directly. This is stated as a named limitation and is reassuring, not a full defence.
- The LeWM gate was tuned on a small diagnostic split and failed to generalise to the full test set. It was reported, not re-tuned.
- Earlier stronger-looking results relying on shortcuts, transductive support or oracle access were corrected rather than reported.

# References cited in this deck


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- Zhou et al. (2025). DINO-WM: World Models on Pre-Trained Visual Features Enable Zero-Shot Planning. ICML.

# Objective achievement

| OBJ. | STATUS | BASIS |
|---|---|---|
| 1 | Fully | Three-branch benchmark implemented as designed and used for every family reported |
| 2 | Partly | Local exposure matched, externally pretrained backbones carry an advantage not equalised |
| 3 | Within tested configurations | Whether and when selection helps determined for every combination, one positive, a likely explanation for each other |

Objectives 4 and 5 are in the Conclusion chapter's objective table. Every non-positive-route explanation is the study's own interpretation, not independently proven.